

· OPENCLAW COURSE · SECURITY MODULE

Complete Guide:

1Password + OpenClaw

Protect your API Keys like a pro

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Simple alternative

· OpenClaw

Beginner's guide · From zero to working

2025

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SECTION 1

Why use 1Password?

Before starting to configure, it's important to understand why use a secrets manager instead of saving the keys in .env files or in the code. The short answer: security, convenience and professionalism.

ASPECT

· .ENV FILE

· 1PASSWORD CLI

Security

· High risk

Easy to leak via git, logs or backups

· Encrypted vault

AES-256, zero-knowledge

Sharing

· Manual

Copy/paste, WhatsApp, Slack...

· Shared Vault

Team-controlled access

Key rotation

· Laborious

Edit files on each machine

· Updates in one place

All projects use automatically

Audit

· No history

Who accessed? Impossible to know

· Access logs

Complete history of who used

Integration with

CI/CD

· Manual variables

· Native integration

GitHub Actions, GitLab CI, etc

..

ATTENTION

.env files with API keys

HAVE ALREADY BEEN RESPONSIBLE FOR THOUSANDS OF DOLLARS IN DAMAGES for inattentive developers · an OpenAI key committed by mistake can generate a huge bill before you realize it.

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SECTION 2

Create your account on 1Password

Step 1 of 2 · Account

1

Access the official website

Open your browser and go to 1password.com/pt. Click on "Try for free" · the Individual plan is enough for the course and costs ~US\$2.99/month after the 14-day trial.

2

Complete the registration and save the Secret Key

After creating the account, 1Password will generate your Emergency Kit · a PDF with your Secret Key. Print or save this PDF in a safe place. Without it, you cannot regain access.

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COMMON MISTAKE

Never save the Emergency Kit in unencrypted cloud services (Google Drive, Dropbox in plain text). Print it physically or use another encrypted vault.

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TIP

The plan

TEAMS OR FAMILIES

allows you to share vaults with project colleagues. If you work in a team, consider these plans from the beginning.

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SECTION 3 AND 4

Install the App and the CLI

Step 2 of 6 · Installation

3.1 · Install the Desktop App

The desktop app is necessary for the integration with the CLI to work (it acts as the local vault that the CLI consults without needing a password every time).

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macOS

Download from

1password.com/downloads or

through the Mac App Store

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Windows

Download the .exe installer from

1password.com/downloads

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Linux

.deb/.rpm package available at

1password.com/downloads/linux

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TIP · LINUX

On Linux, install via the official repository to receive automatic updates.

1Password has native support for Ubuntu, Fedora and Arch.

3.2 · Install the CLI op

The CLI op is the program you will use in the terminal and that OpenClaw will call automatically to search for your credentials.

macOS via Homebrew (recommended)

brew install 1password-cli

Check installation

op --version

BASH · MACOS

Add GPG key from repository

curl -sS <https://downloads.1password.com/linux/keys/1password.asc> | \

sudo gpg --dearmor --output /usr/share/keyrings/1password-archive-keyring.gpg

BASH · LINUX

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ATTENTION · MACOS WITH HOMEBREW

On macOS, Homebrew installs op in /opt/homebrew/bin/op (Apple Silicon) or /usr/local/bin/op (Intel). Note this path · you'll need it in the OpenClaw configuration.

Add apt repository

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/1password-archive-keyring.gpg]
```

```
https://downloads.1password.com/linux/debian/$(dpkg --print-architecture) stable main" | \
```

```
sudo tee /etc/apt/sources.list.d/1password.list
```

Install

```
sudo apt update && sudo apt install 1password-cli
```

Windows via winget

```
winget install AgileBits.1Password.CLI
```

Verify (open a new terminal after installing)

```
op --version
```

POWERSHELL · WINDOWS

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SECTION 5 AND 6

Desktop Integration and First

Commands

Step 3 of 6 · Configuration

5 · Enable Desktop Integration

This step is the secret to prevent the CLI from asking for a password every time · it uses the desktop app as an authenticator. Without this, you will need to log in via terminal every time.

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WINDOWS · PREREQUISITE

On Windows, it is necessary to enable

WINDOWS HELLO

(facial recognition, fingerprint or PIN) before activating integration with the CLI.

1Password · Preferences

..Navigate to

Settings · Developer

.

· Integrate with 1Password CLI

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Integrate with 1Password CLI · enable this toggle

Connect with SSH Agent (optional)

· macOS: Settings · Developer | · Windows: enable Windows Hello first | · Linux:

Settings · Security first

.. ON THE SCREEN YOU WILL SEE

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LINUX · PREREQUISITE

On Linux, first go to

SETTINGS · SECURITY

and enable

"UNLOCK USING SYSTEM AUTHENTICATION"

. Then go to

SETTINGS · DEVELOPER

to integrate the CLI.

6 · First Commands in the Terminal

With the app installed and integration enabled, open the terminal and test:

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SUCCESS

View your active account

op whoami

List your vaults

op vault list

List items in a vault

op item list --vault "Private"

If you are not logged in, explicitly log in

op signin

BASH

Terminal · bash

\$ op whoami

Email: seu@email.com

Account: minha-conta

UserUUID: ABCDEF123...

\$ op vault list

ID NAME

abc123def456... Private

xyz789ghi012... Shared

· CLI working correctly!

.. ON THE SCREEN YOU WILL SEE

If op whoami showed your email and op vault list listed your vaults · congratulations!
The integration is working. The 1Password app authenticates in the background using
biometrics or PIN.

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SECTION 7 AND 8

Create and Read Credentials

Step 4 of 6 · Credentials

7 · Create an API Key Item

Now let's save your Anthropic (or OpenAI) key in 1Password. You can do this through the graphical interface or the CLI.

Option A · Graphical interface (easier for beginners):

Option B · Via CLI (faster if you're already in the terminal):

1Password · Vault Private

1..

Click on

+ New Item

in the upper right corner

2..

Select

· API Credential

from the list of types

3..

Title: Anthropic API Key | Credential field: paste your sk-ant-api03-... key

4..

Click

· Save

.. ON THE SCREEN YOU WILL SEE

Create API credential item in vault Private

op item create \

--category=api-credential \

--title="Anthropic API Key" \

--vault="Private" \

credential=sk-ant-api03-XXXXX

For OpenAI:

op item create \

--category=api-credential \

--title="OpenAI API Key" \

BASH

8 · Read Credentials via CLI

There are two ways to read your credentials · with `op item get` for inspection or with `op read` for use in scripts.

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SECRET REFERENCE URI

The format `op://Vault/Item/Field` is called

SECRET REFERENCE

. It is the safest way to reference credentials · the value is never exposed in the code, just the path to it.

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COMMON ERROR · VAULT WITH SPACES

If your vault is called `My Vault` (with a space), the URI becomes `op://My Vault/Item/Field` . If there is an error, try using the numeric ID of the vault: `op vault list` shows the IDs.

```
--vault="Private" \
```

```
credential=sk-proj-XXXXXX
```

```
# Read specific field (with reveal to see the value)
```

```
op item get "Anthropic API Key" --field credential --reveal
```

```
# Use via URI op:// (Secret Reference format)
```

```
op read "op://Private/Anthropic API Key/credential"
```

```
# Use directly in an environment variable export
```

```
export ANTHROPIC_API_KEY=$(op read "op://Private/Anthropic API  
Key/credential")
```

```
# Confirm it worked
```

```
echo $ANTHROPIC_API_KEY
```

```
BASH
```

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SECTION 9

Integrate with OpenClaw

Step 5 of 6 · Integration

With 1Password working, it's time to connect everything to OpenClaw. OpenClaw supports a system called SecretRef exec · instead of storing the key in openclaw.json , you configure OpenClaw to call the op CLI in real time to fetch the credential.

9.1 · Configuration in openclaw.json

9.2 · macOS with Homebrew (different path)

On macOS with Homebrew (especially Apple Silicon), op is in a different path and is a symlink. Add allowSymlinkCommand :

```
{
  "secrets": {
    "providers": {
      "onpassword": {
        "source": "exec",
        "command": "/usr/local/bin/op",
        "args": ["read", "op://Private/Anthropic API Key/credential"],
        "passEnv": ["HOME"],
        "jsonOnly": false
      }
    }
  },
  "models": {
    "providers": {
      "anthropic": {
        "apiKey": {
          "source": "exec",
          "provider": "onpassword",
          "id": "value"
        }
      }
    }
  }
}
```

JSON · OPENCLAW.JSON

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DISCOVERING THE OP PATH

To discover where op is installed: which op on macOS/Linux. On Windows:

where op in CMD or Get-Command op in PowerShell.

9.3 · Simple Alternative (without SecretRef)

For beginners who want to use 1Password only as a manual vault and configure the environment before starting OpenClaw:

.

TIP · ADD TO YOUR SHELL PROFILE

Add the export above to your ~/.zshrc or ~/.bashrc so that the variable is always loaded when you open the terminal. The key is never saved in the file · op fetches from the vault in real time.

```
{
  "source": "exec",
  "command": "/opt/homebrew/bin/op", // Apple Silicon
  "allowSymlinkCommand": true,
  "trustedDirs": ["/opt/homebrew"],
  "args": ["read", "op://Private/Anthropic API Key/credential"],
  "passEnv": ["HOME"],
  "jsonOnly": false
}
```

JSON · MACOS HOMEBREW

Export the variable using the op CLI

```
export ANTHROPIC_API_KEY=$(op read "op://Private/Anthropic API
Key/credential")
```

Restart the gateway with the key available

```
openclaw gateway restart
```

BASH

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SECTION 10

Full Flow: openclaw secrets

Step 6 of 6 · Activation

OpenClaw has a dedicated command for managing secrets. The complete flow is simple · it's 5 commands in sequence:

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Audit
secrets
audit

.

..

Configure
secrets
configure

.

.

Apply
secrets
apply

.

.

Reload
secrets
reload

.

.

Verify
audit --
check

1. Audit the current state of credentials

Shows which secrets are configured and which are missing

openclaw secrets audit --check

2. Start interactive configuration

Guides you step-by-step for each secret needed

openclaw secrets configure

3. Apply the generated configuration

openclaw secrets apply --from /tmp/openclaw-secrets-plan.json

4. Reload the runtime to use the new secrets

openclaw secrets reload

5. Verify again · it should show everything green

openclaw secrets audit --check

BASH · FULL SEQUENCE

Terminal · openclaw secrets audit --check

\$ openclaw secrets audit --check

Checking secret providers...

· anthropic.apiKey · resolved via exec (onpassword)

· openai.apiKey · resolved via exec (onpassword)

.. ON THE SCREEN YOU WILL SEE

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SUCCESS

When the audit shows all the required secrets as

· RESOLVED

, OpenClaw is ready to use the 1Password credentials automatically ·
without you needing to do anything every time you start it.

.

ERROR · "OP: COMMAND NOT FOUND" OR "EXEC FAILED"

This means that the path to op in openclaw.json is wrong. Confirm with
which op and update the "command" field with the correct full path.

· slack.botToken · not configured (optional)

· All required secrets resolved!

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SECTION 11

Final Checklist

Use this list to confirm that everything is configured correctly before you start using OpenClaw with 1Password:

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Account created on 1Password

You signed up at 1password.com and saved your Emergency Kit in a safe place

.

Desktop app installed and logged in

The app is installed on your OS, you are logged in and can see your vaults

.

CLI installed and verified

op --version returns a valid version (ex: 2.x.x)

.

Desktop integration enabled

Settings · Developer · "Integrate with 1Password CLI" is enabled

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op whoami works without asking for a password

The CLI authenticates via biometrics/PIN from the app, without typing the master password

.

API Keys saved in 1Password

Anthropic, OpenAI or other keys are as "API Credential" items in the Private vault

.

op read works

op read "op://Private/Anthropic API Key/credential" returns the key without error

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openclaw.json configured with SecretRef exec

The command field points to the correct op path (use which op to confirm)

.

openclaw secrets audit --check passes green

All mandatory secrets appear as "resolved"

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OpenClaw gateway working

openclaw gateway status shows running and the agent responds normally

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CONGRATULATIONS!

If all items are checked, you have successfully configured 1Password with OpenClaw. Your API keys are protected in an encrypted vault.

Quick Command Reference

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OpenClaw Course · Security Module

This guide covers the complete configuration of 1Password CLI with OpenClaw.

Questions? Access the course group or the documentation at docs.openclaw.ai

· docs.openclaw.ai

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· Course Group

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· 1password.com

... Verify

op --version # CLI version

op whoami # active account

op vault list # list vaults

... Create

op item create --category=api-credential --title="Name" credential=sk-...

... Read

op read "op://Private/Name/credential"

op item get "Name" --field credential --reveal

... OpenClaw

openclaw secrets audit --check

openclaw secrets configure

openclaw secrets reload

openclaw gateway status

BASH · CHEATSHEET

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